Mohammad Aria Amini

M.Sc. Student in Petroleum Geology

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ABOUT

I am Mohammad Aria Amini, an M.Sc. student in Petroleum Geology and a researcher, writer, and scientific editor in the field of Earth sciences. I am interested in research, scientific writing and editing, and continuously developing my knowledge and professional skills in Earth sciences.

INTERESTS

My research and professional interests primarily lie in the field of Earth sciences, with a particular focus on geochemistry, petroleum geology, and topics related to subsurface resources. I am interested in studying various areas of geology, particularly geological and geochemical processes, rocks and minerals, petroleum systems, and the application of geological data to the understanding and assessment of subsurface resources. I am also interested in applying modern data processing, analytical methods, and computational technologies to Earth science research, and I continuously seek to expand my knowledge and skills in these areas.

EDUCATION

M.Sc. in Petroleum Geology — Oct 2026 – Present
Shahid Beheshti University, Tehran, Iran

B.Sc. in Geology — 2022 – 2026
Golestan University, Faculty of Earth Sciences, Gorgan, Iran

– Ranked first among incoming undergraduate students, with a Grade A average
– Completed the program in seven semesters – one semester ahead of the standard curriculum
– Completed 6 of 7 terms at Shahid Beheshti University, Tehran

High School Diploma, Experimental Sciences — 2017 – 2020
Shahid Beheshti Exceptional Talents High School (NODET), Ahvaz, Iran

PROFESSIONAL EXPERIENCE

Varjavand Intelligent Creative Software & Hardware Services Co.
Chief Executive Officer (CEO) and Chairman of the Board · Full-Time — Jul 2025 – Present
Website: varjavand.org

Arian Zamin Publishing House
Director · Full-Time — Apr 2026 – Present
Editorial Specialist and Publications Researcher · Part-Time — Mar 2023 – Apr 2026

Pars Geological Research Center (Arian Zamin)
Geology Specialist · Full-Time — Feb 2023 – Present

Gemology Science and Art Development Co. (Gemsa), Knowledge-Based Company
Research & Development (R&D) Manager · Full-Time — Sep 2024 – Apr 2025

RESEARCH & INVESTIGATIVE EXPERIENCE

Academic Internship — May 2025 – Aug 2025
National Iranian Oil Company
Internship report: National Iranian South Oilfields Company, Geological Operations Department · Grade A

Research & Development — Sep 2024 – Apr 2025
Gemology Science and Art Development Co. (Gemsa)
Research on Raman spectroscopy using spectrophotometric methods · Supervised by Prof. Fariborz Masoudi (Full Professor)

Research Collaborator and Manager of HPC Infrastructure — Oct 2024 – Present
Institute of Medicinal Plants and Raw Materials, Shahid Beheshti University
Designed, built, and operated the group's computing infrastructure and processing environment · Directly assisted with running and compiling code, and supported its users · Supervised by Dr. Hassan Rezadoost Chahardehi (Associate Professor) · h_rezadoost@sbu.ac.ir

Research Collaborator and Manager, Physical Chemistry Computing (HPC) Core — Apr 2025 – Present
Faculty of Chemistry and Petroleum, Shahid Beheshti University, Tehran
Managing and developing the faculty's HPC infrastructure to support computational and physical chemistry research · Also assisted with managing computing users, compiling code, and running computations · Supervised by Prof. Mansour Zahedi (Full Professor) · m-zahedi@sbu.ac.ir

Earth Sciences and Petroleum Researcher — Jan 2023 – Present
Pars Geological Research Center (Arian Zamin)
Research on economic geology, mineral deposits, and petroleum geology, including petrographic and geochemical data analysis · Supervised by Prof. Mansour Ghorbani (Full Professor) · m-ghorbani@sbu.ac.ir

TEACHING ACTIVITIES

Teaching Assistant (TA) — Mining Engineering Technical Services and Mining Regulations — Feb 2026 – Sep 2026
Shahid Beheshti University, Tehran, Faculty of Earth Sciences
Course for 8th-semester undergraduate Geology students · Supervised by Prof. Mansour Ghorbani (Full Professor) · m-ghorbani@sbu.ac.ir

Teaching Assistant (TA) — Igneous Petrology — Oct 2025 – Feb 2026

Shahid Beheshti University, Tehran, Faculty of Earth Sciences
Course for third-semester undergraduate Geology students · Supervised by Prof. Mansour Ghorbani (Full Professor) · m-ghorbani@sbu.ac.ir

Teaching Assistant (TA) — Igneous Petrology Oct 2024 – Feb 2025
Shahid Beheshti University, Tehran, Faculty of Earth Sciences
Course for third-semester undergraduate Geology students · Supervised by Prof. Mansour Ghorbani (Full Professor) · m-ghorbani@sbu.ac.ir

Research Week Events & Workshops Series Jan 2026 – Feb 2026
Shahid Beheshti University, Tehran
Instructor · Workshop: Introduction to Parallel Processing Systems and Linux-Based High-Performance Computing Environments

PATENTS

System and Method for Depth-Aware Identification and Zonation of Productive (Net Pay) Intervals in Oil Wells Using Multi-Model Probabilistic Fusion of Well-Log Data Registered · 2026
Iranian Patent · Owner and Sole Inventor
Application No. 140450140003008805 · Filed 18 Dec 2025 (1404/09/27) · Registration No. 114471 · Published 23 Sep 2026 (1405/07/01) · IPC: E21B 43/00 · View on Iranian Patent Office · View Patent Registration Certificate
A depth-aware software system that standardizes and quality-controls multi-well log data, builds depth-based features, trains several classifiers, calibrates their probabilities and fuses them into per-depth zone labels with an uncertainty index.

BOOKS

Ghorbani, M., & Amini, M. A. (2025). The Historiography of Earth Science in Iran: Introducing a Number of Contemporary Pioneers.
Co-author · ISBN: 978-600-6058-43-6

Fathi, T. (2026). Petroleum Contamination Hydrogeology: Geological, Environmental, Legal Aspects, and Models for the Remediation of Contaminated Groundwater.
Scientific Editor · ISBN (from the second edition): 978-600-6058-60-3

Momenzadeh, M. (2025). The Mythical City of Zabol: The Burnt City.
Editor · ISBN: 978-600-6058-44-3

Salmati, R. (2025). A Comprehensive Guide to Geological Mapping at Scales of 1:50,000 and Larger.
Editor · ISBN: 978-600-6058-42-9

CONFERENCES

Paper Title: **Natural Resources (Mineral, Energy, and Water Resources) in Iran** 2025
4th Global Summit on Advances in Earth Science & Climate Change (Adv. ESCC 2025) - Berlin, Germany (Virtual Presentation)
Oral Presentation
Authors: Mansour Ghorbani, Mohammad Aria Amini (Speaker) · Distinguished Speaker · Organized by Peers Alley Media · Sep 29–30, 2025

Paper Title: **Differences in Garnet Classification Based on Color** 2024
43rd Earth Sciences Congress (National Conference) - Tehran, Iran
Conference Paper
Authors: Ziba Delpasand, Fariborz Masoudi, Mohammad Aria Amini · 10 pages · Persian · National Sci-Doc ID: GSI43_151 · View on Civilica

Paper Title: **Petrographic, Mineralization, and Alteration Study of the Choudarchay Copper Vein Deposit** 2024
43rd Earth Sciences Congress (National Conference) - Tehran, Iran
Oral Presentation
Author: Mohammad Aria Amini · 13 pages · Persian · National Sci-Doc ID: GSI43_064 · View on Civilica

Paper Title: **Economic Geology of Copper and Associated Elements in the South Rayen Area, with a Focus on Gadar-e-Siah** 2023
42nd Earth Sciences Congress (National Conference) - Tehran, Iran
Oral Presentation
Authors: Mohammad Aria Amini, Shohreh Salkhordeh · 15 pages · Persian · National Sci-Doc ID: GSI42_161 · View on Civilica

HONORS & AWARDS

Top Young Earth Sciences Researcher 2024
Geological Society of Iran
Awarded for the paper "Petrographic, Mineralization, and Alteration Study of the Choudarchay Copper Vein Deposit"

LANGUAGES

Persian – Native | **English** – Professional Working Proficiency | **French** – Learning
GRE General Test Taken: Dec 11, 2024 · Expires: Dec 11, 2029 · Verbal Reasoning 166/170 · Quantitative Reasoning 166/170 · Analytical Writing 4.0/6.0 (63rd percentile)
TOEFL iBT Taken: Jan 1, 2025 · Expires: Jan 1, 2027 · Total Score 102/120 · Reading 30/30 · Listening 27/30 · Writing 28/30 · Speaking 17/30

PROFESSIONAL MEMBERSHIPS

Member of the Iranian Geological Society Oct 2022 – Present
Full (regular) member of the society, and also Executive Secretary and Website Manager of the Iranian Geological Society
Member of the Geological Society of America (GSA)

Member of the American Association of Petroleum Geologists (AAPG)

VOLUNTARY & SOCIAL ACTIVITIES

Volunteer Supporting Mahak Hospital and Rehabilitation Complex & Patients Mar 2023 – Present
MAHAK Charity — The Society to Support Children Suffering from Cancer
Disaster and Humanitarian Relief · MAHAK ID: 020175959 · MAHAK Membership No.: 10704

Organ Donation Ambassador Apr 2023 – Present
Iranian Society of Organ Donation
Health

Organ Donation Volunteer Aug 2019 – Present
Iranian Society of Organ Donation
Disaster and Humanitarian Relief · Membership No.: 3751462 (ehda.ir)

Volunteer Apr 2023 – Present
Iranian Red Crescent Society Innovation Center
Disaster and Humanitarian Relief

Benevolent Member Apr 2023 – Present
Raad Charity
Veteran Support & Physical Disability Support · Membership No.: 421297
Raad Charity is a private charitable organization dedicated to supporting and empowering people with physical disabilities.

TECHNICAL & SPECIALIZED SKILLS

Earth Sciences & Petroleum Geology
Petroleum Geology • Geochemistry • Petrography & Thin-Section Analysis • Geological Mapping • Geological Data Interpretation • Sedimentology & Stratigraphy • Structural Geology & Tectonics

Geoscience-Specific Software & Tools
ArcGIS / ArcGIS Pro • QGIS • Google Earth Engine • ENVI • RockWorks • Petrel

Programming, Computing & Data Analysis
Python • MATLAB • C# • JavaScript • PHP • Bash / Shell Scripting • Data Analysis & Visualization • Geoscientific Data Modeling & Analysis • Large-Scale Data Processing • High-Performance Computing (HPC)

Infrastructure, Systems & Networking
Linux • Linux System Administration • Server Administration • Computer Networking • Network Infrastructure Management • Virtualization & Virtual Machine Management • Data Center Infrastructure • Server & Compute Resource Management

General Software & Productivity Tools
Microsoft Office (Word, Excel, PowerPoint) • Git • GitHub • Command-Line Environments • Technical Documentation

CERTIFICATES, LICENSES & COURSES

Certificates

Vocational Skill Certificate — WordPress CMS Developer Technical and Vocational Training Organization of Iran · Standard ID 7321205300201 · 2025 · 42245009 · View certificate
Vocational Skill Certificate — Goldsmith Technical and Vocational Training Organization of Iran · Standard ID 7313207900301 · 2025 · 42788555 · View certificate
Petroleum Engineering with AI Applications Coursera Specialization · View Specialization Certificate
Introduction to Environmental Science Coursera Specialization · View Specialization Certificate
Google Digital Marketing & E-commerce Professional Certificate Coursera Professional Certificate · View Professional Certificate

Licenses

Varjavand Intelligent Creative Software Co. — Activity License Tehran Provincial Computer Trade Organization · View license
Home Business License — Artificial Intelligence View license
Home Business License — Software System Architecture Design View license
Home Business License — Gold & Jewelry Design View license
Home Business License — Precious & Semi-Precious Gemstone Cutting View license
Home Business License — Software Programming View license
Business License — Software Installation & Setup Services and Related Activities (Legal Entity)

Courses

Petroleum Engineering with AI Applications

Coursera Specialization · 5 courses
– Hydrocarbon Exploration and Production 2025 · EVKUGMGT45RP · View certificate
– From Wellhead to Refinery: Midstream Oil and Gas Processing 2025 · RO8QDU44OUOL · View certificate
– Natural Gas Production and Processing 2025 · IGS15UQQFRV4 · View certificate
– AI & ML Applications in Oil and Gas Industry 2025 · O4Y2QT274ZI3 · View certificate
– Utilities, Safety & Environmental Care in Oil & Gas Industry 2025 · UQWIYV2XAVAV · View certificate
View Specialization Certificate

Introduction to Environmental Science

Coursera Specialization · 3 courses
– Environmental Science 2024 · 67L7TVDOHCS4 · View certificate
– Population, Food, and Soil 2024 · E1UHZEYRKK2T · View certificate
– Energy and Environment 2024 · TAEBBTUG4LV3 · View certificate

View Specialization Certificate

Google Digital Marketing & E-commerce Professional Certificate

Coursera Professional Certificate · 7 courses

– Foundations of Digital Marketing and E-commerce 2024 · ADAC17OAAUTM · View certificate
– Attract and Engage Customers with Digital Marketing 2024 · BYDLT1NC85UY · View certificate
– From Likes to Leads: Interact with Customers Online 2024 · OY3RTJ0FB1O9 · View certificate
– Think Outside the Inbox: Email Marketing 2025 · J8KTD6HZ13TU · View certificate
– Assess for Success: Marketing Analytics and Measurement 2025 · Z4TCNOZKS8MI · View certificate
– Make the Sale: Build, Launch, and Manage E-commerce Stores 2025 · TOTMC7EZJCGQ · View certificate
– Satisfaction Guaranteed: Develop Customer Loyalty Online 2025 · 7IHL2H68QMKE · View certificate

View Professional Certificate

Sustainable Neighborhoods Johns Hopkins University · 2025 · PD3013GV4N42 · View certificate
Sustainable Transportation Networks and Streetscapes Johns Hopkins University · 2025 · BQ9NUFWQ8JHO · View certificate
Wind Energy Technical University of Denmark (DTU) · 2025 · BF3120EB7YPC · View certificate
Sustainable Regional Principles, Planning and Transportation Johns Hopkins University · 2025 · HIHAH8WSFR95 · View certificate
Our Earth: Its Climate, History, and Processes University of Manchester · 2023 · 6Y537YJ5QH9J · View certificate
Extinctions: Past, Present, & Future Emory University · 2023 · EK5WJUBLEB92 · View certificate
Electric Power Systems University at Buffalo · 2023 · QFFR5N2AQ26W · View certificate
Introduction to GIS Mapping University of Toronto · 2023 · RXDSMTEBNYCY · View certificate
Solar Energy Basics The State University of New York · 2023 · PTMSZ9B6KQ8F · View certificate
Oil & Gas Industry Operations and Markets Duke University · 2023 · G7DYSPR7M3RR · View certificate
Unity: Design & Deform Meshes for 3D Geometry Control EDUCBA · 2025 · PAOC4O9D4ALK · View certificate
Positive Psychiatry and Mental Health The University of Sydney · 2025 · LQ39UU674WEF · View certificate
Firm Level Economics: Markets and Allocations University of Illinois Urbana-Champaign · 2024 · XICPQ5ZM9KSV · View certificate
Introduction to Web Development University of California, Davis · 2024 · OISIZ0VHN9AP · View certificate
How to Write and Publish a Scientific Paper École Polytechnique · 2024 · TRG7BMSS8Q9I · View certificate
Firm Level Economics: Consumer and Producer Behavior University of Illinois Urbana-Champaign · 2024 · 1JH025ZXSAFZ · View certificate
Build a Full Website using WordPress Coursera · 2023 · FLH3PRW9CBDC · View certificate
COVID-19 Contact Tracing Johns Hopkins University · 2023 · CTXPE9DBBSWB · View certificate
High Stakes Leadership: Leading in Times of Crisis University of Michigan · 2023 · XWTEBYE99LJU · View certificate